

A Captive-Habitat FTPL Model Embedded in New Keynesian Structure

Technical specification for numerical implementation

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Abstract

This document specifies the *microfounded* version of the captive-demand FTPL model for numerical solution. It replaces the reduced-form regulatory mandate of the Entrega III draft (in which the captive segment was a constraint, the liability pool and regulatory intensity were exogenous, and the wedge fell out of an accounting definition) with a two-agent *preferred-habitat* structure in which the long-bond wedge and the fiscal rent *emerge from optimization*. The portfolio block is embedded in a standard New Keynesian environment under passive monetary policy, exactly as in Entrega III, so the solution method (linearize around a deterministic steady state, Klein/QZ, Blanchard–Kahn count, IRFs) is unchanged in spirit. What changes is the set of equilibrium conditions, the steady state (which must now be computed numerically at a *binding-mandate*, distorted point), and the consequent state/jump classification. This spec is the object the implementation is built on; it is written to be handed to a programming collaborator. Cross-references to the two source documents use the tags [HAB] (the preferred-habitat technical note) and [E3] (Entrega III).

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1 What this model is, and what changed

The dissertation’s claim is that a binding regulatory mandate forces a captive investor segment to hold long-duration sovereign debt at a price an unconstrained investor would not pay; the resulting wedge generates a per-period rent that backs the price level, providing a third margin of fiscal adjustment alongside surpluses and inflation. In Entrega III this was implemented by *positing* a mandate $Q_t^L b_t^L = \lambda_t^L V_t^{\text{liab}}$ with $\lambda_t^L, V_t^{\text{liab}}$ exogenous. The advisor’s objection was that this is “too exogenous”: nobody chooses to pay the rent, the wedge is definitional, and the captive segment is a constraint rather than an agent.

This spec answers that objection by making the captive segment an *optimizing* agent with a *preferred-habitat* portfolio motive. There are two agent types. Type *A* (short-habitat) and type *B* (long-habitat) both optimize over consumption and a portfolio of one short and one long nominal bond, subject to a quadratic penalty for deviating from a preferred long-bond share. The regulator’s mandate now binds on agent *B*’s long-bond holdings and *replaces B’s long-bond Euler equation*; agent *B* is pushed away from its own optimum, agent *A* absorbs the residual supply and is in turn pushed off *its* habitat, and the fiscal rent is the cross-sectional portfolio object this distortion creates. Crucially, the rent and the observable price wedge are now *generically distinct* objects (Section 2.6), which is a genuine result rather than a repair.

Object	Entrega III (reduced form)	This spec (micro-founded)
Captive segment	A constraint on a representative household	An optimizing agent <i>B</i> with habitat θ^B
Wedge ω_t	Accounting definition $1 + \omega_t = Q_t^L / Q_t^{L,\text{fund}}$	Equilibrium outcome of distorted portfolios
Fiscal rent	$R_t = \frac{\omega_t}{1+\omega_t} Q_t^L b_t^L$ (price-wedge revaluation)	X_t , cross-sectional determinant (13)
Pricing kernel	Single representative SDF M_{t+1}	Agent <i>A</i> ’s SDF M_{t+1}^A
Headline identity	$v_t = \mathbb{E}_t \sum \prod M(s + R)$	$v_t = \mathbb{E}_t \sum \prod M^A(s + X)$, (18)
New stationarity issue	none (rep. agent)	<i>A–B</i> wealth distribution (unit root)

Three implementation consequences follow, each handled explicitly below and flagged in red where a decision is required:

1. The A – B relative-wealth distribution is a new state and carries a unit root under equal β, σ ; we add a small wealth-elastic closing device (Section 2.3).
2. The clean closed-form reference point of [HAB] has *zero* wedge and rent, so the steady state must be computed numerically at a deliberately *distorted*, binding-mandate point that hits the empirical wedge target (Section 4).
3. The rent’s sign responses are conditional, not mechanical, so the four sign predictions must be *verified numerically* at the calibrated point (Section 6).

2 The nonlinear model

2.1 Environment

Time is discrete, $t = 0, 1, 2, \dots$. One non-storable good. Agent types $i \in \{A, B\}$ with population shares $\mu_A + \mu_B = 1$. Aggregate resource constraint [HAB EQ. 1]:

$$\mu_A c_t^A + \mu_B c_t^B + g_t = y_t, \quad (1)$$

where y_t is produced by the New Keynesian firm block (Section 2.9) and g_t is government spending. Nominal price level P_t , gross inflation $\pi_t \equiv P_t/P_{t-1}$. Lowercase quantities are real unless stated.

The government issues two nominal instruments [HAB §1]:

- a one-period **short bond**, real stock b_t^S , real price Q_t^S ;
- a **geometric perpetuity** (long bond) with decay $\delta \in [0, 1)$: a unit issued at t pays nominal coupons $1, \delta, \delta^2, \dots$. Real stock (fresh-bond-equivalent) b_t^L , real price Q_t^L .

Maturity composition is exogenous [HAB EQ. 3]:

$$b_t^L = \chi_t (b_t^S + b_t^L), \quad \chi_t \in [0, 1]. \quad (2)$$

Agent i holds $b_t^{S,i}, b_t^{L,i}$ in real per-capita terms. Market clearing [HAB EQ. 4]:

$$\mu_A b_t^{S,A} + \mu_B b_t^{S,B} = b_t^S, \quad \mu_A b_t^{L,A} + \mu_B b_t^{L,B} = b_t^L. \quad (3)$$

2.2 Preferences and habitat cost

Each agent maximizes [HAB EQ. 10–11]

$$\mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \left[\frac{(c_t^i)^{1-\sigma}}{1-\sigma} - H^i(b_t^{S,i}, b_t^{L,i}) \right], \quad H^i = \frac{\phi}{2} (\alpha_t^i - \theta^i)^2 W_t^i, \quad (4)$$

with portfolio wealth and long-share

$$W_t^i \equiv b_t^{S,i} + b_t^{L,i}, \quad \alpha_t^i \equiv \frac{b_t^{L,i}}{W_t^i}, \quad (5)$$

$\phi > 0$, $\theta^i \in [0, 1]$, and the key heterogeneity $\theta^A < \theta^B$ (A short-habitat, B long-habitat). $\beta \in (0, 1)$ and $\sigma > 0$ are common across agents.

Limiting interpretation (useful for sanity-checking the code). As $\phi \rightarrow \infty$ with θ^B fixed at the mandated long share, agent B ’s habitat penalty collapses toward a hard constraint $\alpha^B \rightarrow \theta^B$, i.e. toward the Entrega III reduced-form mandate. The microfounded model nests the reduced form as this limit; finite ϕ is the economically interesting interior.

2.3 Closing device (relative-wealth stationarity)

With equal β, σ and only nominal bonds, the A – B wealth split is indeterminate in levels (a continuum of steady states; a unit root in the linearization). The habitat penalty does *not* cure this — it penalizes portfolio *composition*, not the wealth *level*. We adopt the small wealth-elastic device standard in the open-economy literature (the analogue of Schmitt-Grohé–Uribe’s debt-elastic premium used as the “closing device” in the SOE-FTPL north-star paper). Define agent A ’s normalized wealth deviation $\tilde{W}_t^A \equiv W_t^A - \bar{W}^A$ and let each agent face a small additive premium on its *short-bond* return that is elastic to its own wealth deviation:

$$\text{effective short return wedge for } i: \quad \rho_t^i = -\psi(W_t^i - \bar{W}^i), \quad \psi > 0 \text{ small.} \quad (6)$$

This term enters *only* agent i ’s short-bond Euler equation (10) additively; it vanishes at the reference wealth $\bar{W}^i$ and contributes nothing to steady-state levels. Its sole role is to move the relative-wealth root from exactly 1 to $1 - \epsilon(\psi)$, restoring a unique stationary distribution.

[DECISION 1 — closing-device form.] (6) is one admissible specification (a wealth-elastic return wedge). Two alternatives are equally defensible and equally cheap to code: (i) a convex portfolio-holding cost $\frac{\psi}{2}(W_t^i - \bar{W}^i)^2$ added to the period utility; (ii) a wealth-elastic discount factor. *All three share the one property that matters: a single stationary wealth root $1 - \epsilon$, with the model’s reported results invariant as $\psi \rightarrow 0$.* The implementation should (a) pick one, (b) verify the relative-wealth eigenvalue is $1 - \epsilon$ rather than exactly 1, and (c) report that IRFs are invariant across $\psi \in \{10^{-3}, 10^{-4}, 10^{-5}\}$. Recommend specification (i) (utility cost) as the least intrusive on the pricing equations.

2.4 Household budget and first-order conditions

Agent i ’s real flow budget constraint [HAB EQ. 14]:

$$c_t^i + Q_t^S b_t^{S,i} + Q_t^L b_t^{L,i} = y_t^i - \tau_t + \frac{b_{t-1}^{S,i}}{\pi_t} + \frac{(1 + \delta Q_t^L) b_{t-1}^{L,i}}{\pi_t}, \quad (7)$$

where y_t^i is agent i ’s real income (labor income plus its share of firm profits; see Section 2.9 and [DECISION 2]) and τ_t is a lump-sum tax. Define the SDF and the habitat deviation [HAB EQ. 16, 19]:

$$M_{t+1}^i \equiv \beta \left(\frac{c_{t+1}^i}{c_t^i} \right)^{-\sigma}, \quad \varepsilon_t^i \equiv \alpha_t^i - \theta^i. \quad (8)$$

The habitat penalty drives a wedge into each Euler equation. Using the exact derivatives [HAB EQ. 20–23], define the leading-order wedge objects

$$\xi_t^{S,i} \equiv \frac{\phi \varepsilon_t^i \alpha_t^i}{(c_t^i)^{-\sigma}}, \quad \xi_t^{L,i} \equiv \frac{\phi(\theta^i - \alpha_t^i)(1 - \alpha_t^i)}{(c_t^i)^{-\sigma}}, \quad (9)$$

with exact second-order residuals $\mathcal{R}_t^{S,i} = \mathcal{R}_t^{L,i} = -\frac{\phi}{2}(\varepsilon_t^i)^2 / (c_t^i)^{-\sigma} = O((\varepsilon_t^i)^2)$. The Euler equations are then [HAB EQ. 29–30]:

$$Q_t^S = \mathbb{E}_t \left[\frac{M_{t+1}^i}{\pi_{t+1}} \right] + \xi_t^{S,i} + \mathcal{R}_t^{S,i} + \rho_t^i, \quad (10)$$

$$Q_t^L = \mathbb{E}_t \left[\frac{M_{t+1}^i (1 + \delta Q_{t+1}^L)}{\pi_{t+1}} \right] + \xi_t^{L,i} + \mathcal{R}_t^{L,i}. \quad (11)$$

(The closing-device term ρ_t^i from (6) appears only in the short Euler.) At leading order the residuals $\mathcal{R}$ are dropped; final identities re-insert them where they matter.

2.5 The regulatory mandate

The regulator imposes a *market-value* mandate on agent B 's long position [HAB EQ. 48–52]:

$$Q_t^L \mu_B b_t^{L,B} \geq m_t, \quad m_t \equiv \lambda_t^L V_t^{\text{liab}}. \quad (12)$$

λ_t^L is regulatory intensity and V_t^{liab} the captive liability pool; both follow exogenous AR(1) processes (Section 2.10). **When the mandate binds it holds with equality and replaces agent B 's long-bond Euler equation (11).** Agent B still prices the short bond via (10). Agent A prices *both* bonds via (10)–(11); this is why the government is priced by A 's SDF.

Why the asymmetry. Under a binding mandate, B is forced onto its long position by regulation, so B 's long-bond Euler is slack (the multiplier on the mandate absorbs the gap). A is unconstrained and is the marginal price-setter for the long bond. The mandate is the only source of departure from frictionless pricing.

2.6 Fiscal rent and the price wedge — distinct objects

The **leading-order fiscal rent** is the wedge term entering the government present-value identity. After substituting the wedge definitions, applying market clearing, and collecting terms [HAB EQ. 68–73, 102]:

$$X_t = \frac{\phi(\alpha_t^A - \theta^A) \mu_B}{(c_t^A)^{-\sigma} W_t^A} \left[b_t^{L,A} b_t^{S,B} - b_t^{S,A} b_t^{L,B} \right] \quad (13)$$

This is a flow-like, cross-sectional *determinant* object built from agent A 's Euler distortion and the portfolio heterogeneity between A and B . It is exact within the leading-order wedge approximation, with second-order residual $\mathcal{R}_t^A = -\frac{\phi}{2}(\alpha_t^A - \theta^A)^2 (c_t^A)^\sigma (b_t^S + b_t^L)$ [HAB EQ. 96].

Separately, the **price-wedge stock revaluation** [HAB EQ. 74–76] uses the fundamental (unconstrained) long-bond price $Q_t^{L,\text{fund}} \equiv Q_t^{L,\text{unc}}$ and $1 + \omega_t \equiv Q_t^L / Q_t^{L,\text{fund}}$:

$$\Psi_t \equiv (Q_t^L - Q_t^{L,\text{fund}}) b_t^L = \frac{\omega_t}{1 + \omega_t} Q_t^L b_t^L. \quad (14)$$

Ψ_t is the Entrega III rent object. X_t and Ψ_t are **generically distinct** [HAB §13]: X_t is the Euler-wedge object that enters the present-value identity; Ψ_t is a stock revaluation of aggregate long-bond supply. The dissertation's sharpened contribution is exactly this: the convenience-yield object the macro-finance literature measures (Ψ_t / ω_t) is *not* the fiscal-backing rent (X_t), and the model says how they relate. **Both must be tracked separately in the code;** ω_t remains the empirical calibration target, while X_t is what enters the valuation identity.

2.7 Government and the augmented FTPL identity

Government flow constraint [HAB EQ. 77], with primary surplus $s_t \equiv \tau_t - g_t$:

$$\frac{b_{t-1}^S + (1 + \delta Q_t^L) b_{t-1}^L}{\pi_t} = s_t + Q_t^S b_t^S + Q_t^L b_t^L. \quad (15)$$

Define the real market value of inherited debt

$$v_t \equiv \frac{b_{t-1}^S + (1 + \delta Q_t^L) b_{t-1}^L}{\pi_t}. \quad (16)$$

Pricing the government's obligations with agent A 's SDF and using A 's Euler equations with the wedge yields the recursive identity [HAB EQ. 85]:

$$v_t = s_t + X_t + \mathcal{R}_t^A + \mathbb{E}_t[M_{t+1}^A v_{t+1}]. \quad (17)$$

Forward iteration with the transversality condition (Section 2.8) gives the **augmented FTPL identity** [HAB EQ. 103], and its leading-order form [HAB EQ. 105]:

$$v_t = \mathbb{E}_t \sum_{j=0}^{\infty} \left(\prod_{k=1}^j M_{t+k}^A \right) (s_{t+j} + X_{t+j} + \mathcal{R}_{t+j}^A), \quad (18)$$

$$v_t = \mathbb{E}_t \sum_{j=0}^{\infty} \left(\prod_{k=1}^j M_{t+k}^A \right) (s_{t+j} + X_{t+j}) + O(\varepsilon^2). \quad (19)$$

The fiscal rent X_t now sits beside the primary surplus as a component of fiscal backing, *micro-founded* rather than posited.

2.8 Transversality and admissibility

The present-value identity requires the terminal term to vanish [HAB EQ. 97–98]. Agent A 's household transversality controls the discounted value of total government debt. For agent B , the bounded-mandate condition $m_t/y_t \leq \bar{M}$ controls the *market value* of mandated long-bond holdings but not short positions; hence an admissibility restriction on bounded gross positions (or natural debt limits) is required [HAB ASSUMPTION 2, §11]. **For the linearized solution this is not binding** — it matters only for the global/nonlinear argument — but it should be stated in the dissertation text and the code should verify the mandate binds with margin (Section 4.4) so the linear-RE assumption is internally valid.

2.9 New Keynesian block

The portfolio block above is embedded in the same minimal NK environment as Entrega III [E3 §5.3.2]: a continuum of monopolistically competitive firms with Calvo pricing yielding a Phillips curve, and a passive Taylor rule. Output is produced (the model is *not* an endowment economy, unlike the bare [HAB] note), so there is an output gap.

$$\text{NKPC:} \quad \hat{\pi}_t = \beta \mathbb{E}_t \hat{\pi}_{t+1} + \kappa \hat{y}_t, \quad \kappa = \frac{(1 - \theta_C)(1 - \beta \theta_C)\sigma}{\theta_C}, \quad (20)$$

$$\text{Taylor rule:} \quad \hat{i}_t = \phi_\pi \hat{\pi}_t + u_t^i, \quad \phi_\pi < 1 \text{ (passive)}, \quad (21)$$

$$\text{Resource:} \quad \bar{c} \hat{c}_t + \bar{g} \hat{g}_t = \bar{y} \hat{y}_t, \quad \hat{c}_t \equiv (\text{aggregate, see DECISION 2}). \quad (22)$$

Here θ_C is the Calvo non-reoptimization probability (renamed from the habitat θ^i to avoid collision), $\hat{i}_t$ the net nominal short rate ($\hat{i}_t = -\hat{Q}_t^S$ to first order), and u_t^i an AR(1) monetary shock.

[DECISION 2 — consumption aggregation / which Euler is the IS curve.] With two optimizing agents we must specify how household demand maps to the single good's IS curve. Two clean options:

1. **Marginal-agent IS (recommended baseline).** Since agent A is unconstrained and prices the bonds, and the government is priced by M^A , use *agent A's* short-bond Euler (10) as the IS curve. Aggregate $\hat{c}_t$ in (22) is then a fixed-weight combination $\hat{c}_t = s_A^c \hat{c}_t^A + s_B^c \hat{c}_t^B$ with steady-state consumption shares s_i^c , and B 's consumption is pinned residually by B 's budget (7) plus the binding mandate. Simplest; keeps the macro block close to Entrega III.
2. **TANK aggregation.** Carry both agents' consumption Euler equations and aggregate explicitly. More standard in the two-agent NK literature but heavier, and the wedge $\xi^{S,i}$ enters each agent's short Euler so the aggregation is not the textbook TANK one.

Recommend option 1 for the first working version; switch to option 2 only if a referee pushes on distributional dynamics. State the choice explicitly in the dissertation.

2.10 Exogenous processes

Six structural shocks, each AR(1) in logs (or in levels for variables admitting zero), $\log(z_t/\bar{z}) = \rho_z \log(z_{t-1}/\bar{z}) + \sigma_z \varepsilon_t^z$, for

$$z \in \{\lambda_t^L, V_t^{\text{liab}}, \chi_t, g_t, \tau_t, u_t^i\}.$$

The headline experiment is a unit innovation to regulatory intensity $\hat{\lambda}_t^L$.

3 Equilibrium conditions (nonlinear system to be linearized)

Collecting the above, a **binding-mandate** rational-expectations equilibrium is a set of processes satisfying, each period:

1. Agent A short Euler (10) and long Euler (11) ($i = A$);
2. Agent B short Euler (10) ($i = B$);
3. **Binding mandate** (12) with equality, *in place of* agent B 's long Euler;
4. Both agents' budget constraints (7);
5. Long-share definitions $\alpha_t^i = b_t^{L,i}/W_t^i$, $W_t^i = b_t^{S,i} + b_t^{L,i}$, and wedge definitions (9);
6. Market clearing (3) and maturity composition $b_t^L = \chi_t(b_t^S + b_t^L)$;
7. Fundamental long price $Q_t^{L,\text{fund}} = \mathbb{E}_t[M_{t+1}^A(1 + \delta Q_{t+1}^{L,\text{fund}})/\pi_{t+1}]$ and wedge $1 + \omega_t = Q_t^L/Q_t^{L,\text{fund}}$;
8. Fiscal rent (13) and (for diagnostics) Ψ_t (14);
9. Government flow constraint (15); v_t definition (16); augmented identity in recursive form (17);
10. NKPC (20), Taylor rule (21), resource constraint (22);
11. Closing device (6); exogenous AR(1) processes (Section 2.10).

Slack-mandate regime (do not implement first). If the mandate were slack, item 3 reverts to agent B 's long Euler (11) and the wedge collapses ($\omega_t \rightarrow 0$, $X_t \rightarrow 0$). The headline analysis lives entirely in the binding regime; the slack regime is relevant only for the occasionally-binding extension, which is *out of scope* for the linear solution (Section 7).

4 Deterministic steady state (numerical)

This is the load-bearing computation: everything downstream (linearization, eigenvalues, IRFs) is defined relative to this point. It *must* be a binding-mandate, distorted point, *not* the closed-form symmetric reference of [HAB §6, §10].

4.1 Why not the closed-form reference

At the symmetric-supply reference $\chi = \bar{\theta}$ (with $\bar{\theta} \equiv \omega_A \theta^A + \omega_B \theta^B$), each agent sits at its habitat: $\alpha^A = \theta^A$, $\alpha^B = \theta^B$ [HAB EQ. 88–89]. Then (13) gives $X = 0$ and the price wedge $\bar{\omega} = 0$. This is the unique point where the mechanism is switched off. Linearizing there would encode a model in which the rent channel is locally absent, and the empirical anchor $\bar{\omega} = 0.14$ is unreachable. **We therefore choose $\chi \neq \bar{\theta}$ on purpose**, so the mandate forces $\alpha^A < \theta^A$, giving $\bar{X} > 0$ and $\bar{\omega} > 0$.

4.2 The steady-state system

At a deterministic steady state with $\pi = \bar{\pi}$ (target $\bar{\pi} = 1$), constant exogenous $\bar{\lambda}^L, \bar{V}^{\text{liab}}, \bar{\chi}, \bar{g}, \bar{\tau}$, the closing-device terms vanish ($\rho^i = 0$). The unknowns are

$$\{\bar{c}^A, \bar{c}^B, \bar{b}^{S,A}, \bar{b}^{L,A}, \bar{b}^{S,B}, \bar{b}^{L,B}, \bar{Q}^S, \bar{Q}^L, \bar{Q}^{L,\text{fund}}\}$$

(plus aggregates $\bar{b}^S, \bar{b}^L, \bar{y}, \bar{v}$ recovered by identities). The defining equations:

$$A \text{ short Euler: } \bar{Q}^S = \beta/\bar{\pi} + \bar{\xi}^{S,A}, \quad (23)$$

$$A \text{ long Euler: } \bar{Q}^L = \frac{\beta(1 + \delta\bar{Q}^L)}{\bar{\pi}} + \bar{\xi}^{L,A} \implies \bar{Q}^L = \frac{\beta/\bar{\pi} + \bar{\xi}^{L,A}}{1 - \beta\delta/\bar{\pi}}, \quad (24)$$

$$B \text{ short Euler: } \bar{Q}^S = \beta/\bar{\pi} + \bar{\xi}^{S,B}, \quad (25)$$

$$\text{Binding mandate: } \bar{Q}^L \mu_B \bar{b}^{L,B} = \bar{\lambda}^L \bar{V}^{\text{liab}}, \quad (26)$$

$$A, B \text{ budgets: } (7) \text{ at steady state, both agents,} \quad (27)$$

$$\text{Market clearing: } \mu_A \bar{b}^{S,A} + \mu_B \bar{b}^{S,B} = \bar{b}^S, \quad \mu_A \bar{b}^{L,A} + \mu_B \bar{b}^{L,B} = \bar{b}^L, \quad (28)$$

$$\text{Maturity comp.: } \bar{b}^L = \bar{\chi}(\bar{b}^S + \bar{b}^L), \quad (29)$$

$$\text{Fundamental price: } \bar{Q}^{L,\text{fund}} = \frac{\beta/\bar{\pi}}{1 - \beta\delta/\bar{\pi}} \quad (\text{i.e. } \frac{\beta}{1-\beta\delta} \text{ at } \bar{\pi} = 1), \quad (30)$$

$$\text{Wedge: } 1 + \bar{\omega} = \bar{Q}^L / \bar{Q}^{L,\text{fund}}, \quad (31)$$

with $\bar{\xi}^{S,i}, \bar{\xi}^{L,i}$ from (9) evaluated at steady-state allocations. Note the two short-Euler equations *jointly* require $\bar{\xi}^{S,A} = \bar{\xi}^{S,B}$ (common $\bar{Q}^S$), an equilibrium restriction across the two agents' distortions.

4.3 Calibration strategy: invert for ϕ at the wedge target

The empirical anchor is $\bar{\omega} = 0.14$ (Entrega III §5.2, four-year UK mean; sensitivity band [0.12, 0.21]). The procedure:

1. Fix conventional parameters: $\beta = 0.99$, $\sigma = 1$, $\delta = 0.96$, $\theta_C = 0.75$, $\phi_\pi = 0.5$, $\bar{\pi} = 1$, $\rho_z = 0.7$ (provisional), population shares μ_A, μ_B , habitats $\theta^A < \theta^B$, and the long-share target $\bar{\chi}$ (Entrega III uses $\bar{\chi} = 0.353$).
2. Treat the supply–demand gap (via $\bar{\chi}$ relative to $\bar{\theta}$) and the habitat strength ϕ as the levers that set the size of the distortion. **Invert numerically for the ϕ (given the gap, or jointly with the mandate tightness $\bar{\lambda}^L \bar{V}^{\text{liab}}$) that delivers $\bar{\omega} = 0.14$.**

3. Read off $\bar{\alpha}^A, \bar{X}, \bar{\Psi}$ and all steady-state ratios needed for the reduced-form coefficients in Section 5.

Existence is essentially free; realism is the live question. An active-distortion steady state exists for *almost all* parameters — the undistorted case $\chi = \bar{\theta}$ is the measure-zero exception. The real question is whether the ϕ that hits $\bar{\omega} = 0.14$ is plausible. *Every outcome is reportable*: (a) plausible $\phi \Rightarrow$ the microfoundation rationalizes the wedge; (b) implausibly large $\phi \Rightarrow$ a *finding* that a 14% wedge is hard to generate from optimizing habitat demand, implying the reduced form was baking in a larger distortion than micro-behavior supports; (c) intermediate $\Rightarrow$ report a band, as Entrega III already does for $[0.12, 0.21]$. There is no pass/fail gate here.

4.4 Sanity checks the solver must pass

1. **Interior allocations:** $\bar{c}^i > 0, \bar{Q}^S, \bar{Q}^L > 0, \alpha^i \in (0, 1), W^i > 0$ [HAB ASSUMPTION 1].
2. **Wedge positive and Assumption 1 binds:** $\bar{\omega} > 0$.
3. **Mandate tightness interior:** the implied regulatory intensity lies strictly inside its admissible range; $\bar{m}/\bar{y} \leq \bar{M}$ [HAB ASSUMPTION 3].
4. **Long-bond pricing condition:** $\beta\delta/\bar{\pi} < 1$.
5. **Mandate binds with margin:** the unconstrained allocation would violate (12), and the binding allocation sits far enough inside that the IRF shocks of Section 6 do not flip it slack (else the linear solution is invalid; see Section 7).
6. **Determinant sign:** $b^{L,A}b^{S,B} - b^{S,A}b^{L,B}$ is the object multiplying $(\alpha^A - \theta^A)$ in (13); verify its sign at the calibrated point (it is negative near the symmetric reference, [HAB EQ. 90–91]) so that the sign of $\bar{X}$ is understood.

5 Log-linearized system

Linearize all equilibrium conditions around the binding-mandate steady state. Hats denote log-deviations (level deviations for s_t and any variable admitting zero). The structure mirrors Entrega III Table 1, with the captive block replaced by the habitat block. Reduced-form coefficients ($\eta, w_S, w_L, \bar{s}/\bar{v}, \bar{X}/\bar{v}$, rent and wedge elasticities, the determinant elasticities) are computed from the steady state of Section 4 and reported by the solver.

#	Linearized equation (schematic)	Role
1	$\hat{c}_t^A = \mathbb{E}_t \hat{c}_{t+1}^A - \sigma^{-1}(\hat{v}_t - \mathbb{E}_t \hat{\pi}_{t+1}) + \zeta^S \hat{\xi}_t^{S,A} - \psi \tilde{W}_t^A$	IS / A short Euler
2	$\hat{\pi}_t = \beta \mathbb{E}_t \hat{\pi}_{t+1} + \kappa \hat{y}_t$	NKPC
3	$\hat{v}_t = \phi_\pi \hat{\pi}_t + u_t^i$	Passive Taylor rule
4	$\hat{Q}_t^{L,\text{fund}} = -\sigma(\mathbb{E}_t \hat{c}_{t+1}^A - \hat{c}_t^A) - \mathbb{E}_t \hat{\pi}_{t+1} + \eta \mathbb{E}_t \hat{Q}_{t+1}^{L,\text{fund}}$	Fundamental long price
5	$\hat{Q}_t^L = \hat{Q}_t^{L,\text{fund}} + \hat{\omega}_t$	Wedge / A long Euler
6	$\hat{Q}_t^S = -\hat{v}_t; \hat{\xi}_t^{S,B} = \hat{\xi}_t^{S,A} \text{ (common } \hat{Q}_t^S)$	B short Euler tie
7	$\hat{Q}_t^L + \hat{b}_t^{L,B} = \hat{\lambda}_t^L + \hat{V}_t^{\text{liab}}$	Binding mandate (replaces B long Euler)
8	$\hat{\alpha}_t^i = \hat{b}_t^{L,i} - \hat{W}_t^i, \hat{W}_t^i = \omega_t^S \hat{b}_t^{S,i} + \omega_t^L \hat{b}_t^{L,i}$	Shares / wealth, $i = A, B$
9	$\hat{\xi}_t^{S,i}, \hat{\xi}_t^{L,i}$ linear in $(\hat{\alpha}_t^i, \hat{c}_t^i)$ via (9)	Habitat wedges
10	$\mu_A \omega_A^{bS} \hat{b}_t^{S,A} + \mu_B \omega_B^{bS} \hat{b}_t^{S,B} = \hat{b}_t^S \text{ (and } L \text{ analogue)}$	Market clearing
11	$\hat{b}_t^L - \hat{b}_t^S = \hat{\chi}_t / [\bar{\chi}(1 - \bar{\chi})]$	Maturity composition
12	$\hat{X}_t = \hat{\alpha}_t^A \cdot \frac{\alpha^A}{\alpha^A - \theta^A} + (\text{determinant elasticities in } \hat{b}_t^{\cdot}) - \sigma \hat{c}_t^A - \hat{W}_t^A$	Fiscal rent (13)
13	$\hat{v}_t = w_S(\hat{b}_{t-1}^S - \hat{\pi}_t) + w_L(\hat{b}_{t-1}^L + \eta \hat{Q}_t^L - \hat{\pi}_t)$	Real debt market value
14	$\hat{v}_t = (\bar{s}/\bar{v})\hat{s}_t + (\bar{X}/\bar{v})\hat{X}_t + \beta \mathbb{E}_t \hat{v}_{t+1} - \beta(\hat{v}_t - \mathbb{E}_t \hat{\pi}_{t+1})$	Augmented FTPL identity (17)
15	$\bar{c} \hat{c}_t + \bar{g} \hat{g}_t = \bar{y} \hat{y}_t; \hat{c}_t = s_A^c \hat{c}_t^A + s_B^c \hat{c}_t^B$	Resource / aggregation
16	$\hat{s}_t = (\bar{\tau}/\bar{y})\hat{\tau}_t - (\bar{g}/\bar{y})\hat{g}_t$	Primary surplus
17	$\tilde{W}_t^A$ law of motion from A 's budget + closing device	Relative-wealth state
18	AR(1): $\hat{z}_t = \rho_z \hat{z}_{t-1} + \varepsilon_t^z, z \in \{\lambda^L, V^{\text{liab}}, \chi, g, \tau, u^i\}$	Exogenous drivers

Notes on coefficients (all from the SS solve):

- $\eta = \delta \bar{Q}^L / (1 + \delta \bar{Q}^L)$ (long-bond duration coefficient; ≈ 0.95 at the Entrega III calibration).
- $w_S = \bar{b}^S / (\bar{v}\bar{\pi})$, $w_L = (1 + \delta \bar{Q}^L) \bar{b}^L / (\bar{v}\bar{\pi})$; the coefficient on $\hat{\pi}_t$ in row 13 is $-(w_S + w_L) = -1$ (one-for-one inflation revaluation of real debt — this should hold *exactly* as a code check, exactly as in Entrega III).
- Row 12 coefficients: the front factor $(\alpha^A - \theta^A)^{-1} \alpha^A$ *diverges* as the distortion vanishes — another reason the SS must sit at $\bar{\alpha}^A \neq \theta^A$. The determinant elasticities multiply the four $\hat{b}_t^{\cdot}$ and are computed from $\partial X / \partial b$ at the SS.
- ρ^i /closing-device coefficient ψ enters rows 1 and 17 only; set small.

5.1 State/jump classification and the Blanchard–Kahn target

Predetermined states: lagged bond stocks $\hat{b}_{t-1}^S, \hat{b}_{t-1}^L$ (or the agent-level lags, depending on how clearing is substituted), the relative-wealth state $\tilde{W}_{t-1}^A$, and the six AR(1) processes. Forward-looking jumps: $\hat{\pi}_t, \hat{c}_t^A, \hat{Q}_t^L, \hat{Q}_t^{L,\text{fund}}, \hat{\omega}_t$, and the portfolio jumps not eliminated by clearing/mandate. **The BK condition requires $\#(\text{unstable eigenvalues}) = \#(\text{jumps})$.** Report the eigenvalue table exactly as Entrega III Table 2, now *including* the relative-wealth root, which the closing device should place at $1 - \epsilon$ (just inside the unit circle). Verify the count is satisfied in the interior of $\phi_\pi \in [0, 1)$ and fails at $\phi_\pi = 1$ (the Leeper regime boundary, Entrega III §5.4.3).

[DECISION 3 — which variables to substitute out.] As in Entrega III §5.4.2 fn. 4, the system can be written compactly by eliminating current bond stocks via market clearing (3) and the maturity rule, reducing the jump dimension. Recommend building *both* the compact form and a redundant full form, and checking they produce identical IRFs (a standard correctness test).

6 Solution, determinacy, and impulse responses

6.1 Solver

Solve with the Klein (1999) generalized-Schur (QZ) method, identical to Entrega III. Inputs: the two coefficient matrices A_0, A_1 of the linear system $A_0 \mathbb{E}_t w_{t+1} = A_1 w_t$ (plus shock loadings). Outputs: policy function (decision rule) and transition matrix. **This is a drop-in reuse of the Entrega III solver**; only the matrices change.

6.2 Determinacy, the doc-2 way

The north-star SOE-FTPL paper obtains analytic determinacy by exploiting *block-triangular* structure (the tradable block isolates; the real exchange rate has a benign backward root; a small core remains for root-counting). **Attempt the analogue here:**

1. The closed-off relative-wealth block (with the closing device) should be block-recursive — its root is $1 - \epsilon$ regardless of the rest. Verify it is block-triangular so determinacy of the rest is *conditional-clean*.
2. The fundamental-pricing block (row 4) is forward-recursive in $\hat{Q}^{L, \text{fund}}$.
3. Aim to reduce the determinacy question to a small core in $\{\hat{\pi}_t, \hat{v}_t, \hat{\omega}_t\}$ where Woodford-style root-counting conditions can be stated *analytically* (as in doc-2 Appendix C), rather than only reporting QZ eigenvalues. Even partial analytic results here are the “up to doc-2 level” rigor the dissertation is targeting.

6.3 Headline experiment and sign predictions

Unit positive innovation to $\hat{\lambda}_t^L$ (a tightening of the matching / LDI mandate), persistence ρ_λ , 40-quarter horizon. The four sign predictions inherited from Entrega III: on impact inflation falls, the long-bond price rises, the rent rises, real debt market value rises; the nominal short rate falls (passive Taylor response). **These are now conditional, not mechanical**, and must be verified numerically:

- B’s physical long position rises iff the strengthened condition $m_t/Q_t^L > \mu_B b_t^{L,B, \text{unc}}$ holds [HAB EQ. 56–57, 109]; locally $\partial b_t^{L,B}/\partial m_t > 0$.
- A’s long-bond *share* falls iff $(1 - \alpha_t^A) \partial b_t^{L,A}/\partial m_t < \alpha_t^A \partial b_t^{S,A}/\partial m_t$ [HAB EQ. 65, 111], with simple sufficient condition $\partial b_t^{S,A}/\partial m_t \geq 0$ [HAB EQ. 66, 112].
- Local positivity of the fiscal rent $X_t > 0$ holds *near the symmetric reference* under Proposition 1 [HAB PROP. 1]; away from it, confirm the sign at the calibrated point.

6.4 Two regime diagnostics (retain from Entrega III)

Both are predictions of the passive-monetary / active-fiscal regime and should reproduce qualitatively: (i) a *contractionary* monetary shock raises inflation on impact (inverted sign vs. active-NK); (ii) the sensitivity of $|\hat{\pi}_0|$ to a $\hat{\lambda}^L$ shock is non-monotone in ϕ_π across $[0, 1]$ and determinacy fails at $\phi_\pi = 1$. These are the diagnostic signs that distinguish the regime [E3 §5.5.2].

6.5 Counterfactual that isolates the mechanism

The limit $\phi \rightarrow 0$ (habitat strength to zero) shuts the rent channel: wedges vanish, $X_t \rightarrow 0$, and a $\hat{\lambda}^L$ shock has no equilibrium effect. Comparing the headline IRF to this counterfactual isolates the contribution of the habitat/captive block — the microfounded analogue of Entrega III's $\bar{\omega} \rightarrow 0$ counterfactual.

7 Implementation notes and the three landmines

1. **Relative-wealth unit root (LANDMINE 1).** Without the closing device the linear system has a root exactly at 1 and the BK count is knife-edge/ill-posed. *Resolution:* closing device (6), root at $1 - \epsilon$. *Code check:* confirm the eigenvalue and report IRF invariance across small ψ .
2. **Expansion point (LANDMINE 2).** The closed-form reference has $\bar{X} = \bar{\omega} = 0$; linearizing there switches off the mechanism and breaks the UK anchor. *Resolution:* numerical binding-mandate SS at $\chi \neq \bar{\theta}$ targeting $\bar{\omega} = 0.14$ (Section 4). *Code check:* $\bar{\omega} > 0$, $\bar{\alpha}^A \neq \theta^A$, row-12 front factor finite.
3. **Conditional signs (LANDMINE 3).** The four sign predictions are not automatic in the microfounded model. *Resolution:* verify numerically at the calibrated point, and/or restrict exposition to the Proposition 1 neighborhood. *Code check:* print the $\partial b / \partial m$ conditions [HAB EQ. 56–67] at the SS.
4. **Occasionally-binding mandate (OUT OF SCOPE for v1).** Linear RE assumes the same equations every period. If the mandate flips slack under a shock, the equation set changes and a piecewise/nonlinear method (OccBin-style) is required. *Resolution for v1:* set the SS so the mandate binds with margin and verify IRF shocks never breach it (Section 4.4 item 5). State the occasionally-binding case as future work.

Suggested toolchain

- **Dynare** (MATLAB/Octave) is the path of least resistance: write the nonlinear model in a `.mod` file, let Dynare compute the steady state (feed it the numerical solver of Section 4 via a `steady_state_model` block or an external solver), and use `check/stoch_simul` for BK diagnostics and IRFs. The binding mandate is imposed by writing the binding-regime equations directly (not as a complementarity).
- **Python** (reusing the Entrega III Klein/QZ code) is fine and keeps everything in one language: `scipy.linalg.ordqz` for the generalized Schur, plus a root-finder (`scipy.optimize.root`) for the steady state and the ϕ -inversion. Recommended if the Entrega III pipeline is already in Python.

Recommended build order

1. Steady-state solver (Section 4); pass all checks in Section 4.4.
2. ϕ -inversion to hit $\bar{\omega} = 0.14$; report the implied ϕ and judge realism.
3. Linearization (Section 5); verify the -1 inflation coefficient in row 13 and the relative-wealth root at $1 - \epsilon$.
4. Klein/QZ solve; reproduce the BK count and the $\phi_\pi = 1$ boundary.
5. Headline IRF to $\hat{\lambda}^L$; verify the four signs and the two regime diagnostics.
6. Counterfactual $\phi \rightarrow 0$; sensitivity over $\bar{\omega} \in [0.12, 0.21]$, ρ_λ , $\phi_\pi \in [0, 0.95]$, ψ .

Appendix: notation map to source documents

Symbol	Meaning	Source
A, B	short-/long-habitat agent types	[HAB §3]
$\theta^A < \theta^B$	preferred long-bond shares (habitats)	[HAB EQ. 13]
ϕ	habitat penalty strength	[HAB EQ. 11]
$\alpha_t^i, \varepsilon_t^i$	long-share and deviation $\alpha^i - \theta^i$	[HAB EQ. 7,19]
$\xi_t^{S,i}, \xi_t^{L,i}$	short/long habitat wedges	[HAB EQ. 22–23]
M_{t+1}^A	agent A 's SDF (prices government debt)	[HAB EQ. 16]
X_t	fiscal rent (cross-sectional determinant)	[HAB EQ. 102]
Ψ_t, ω_t	price-wedge revaluation; price wedge	[HAB EQ. 75–76]
v_t	real market value of inherited debt	[HAB EQ. 78]; [E3 EQ. 28]
$m_t = \lambda_t^L V_t^{\text{liab}}$	mandate target	[HAB EQ. 49]; [E3 EQ. 22]
δ, η	perpetuity decay; duration coefficient	[HAB §1]; [E3 TAB. 3]
$\phi_\pi < 1, \theta_C, \kappa$	passive Taylor coeff.; Calvo prob.; NKPC slope	[E3 EQ. 43–44]
ψ	closing-device elasticity (new)	this spec
$\bar{\omega} = 0.14$	UK wedge calibration target	[E3 §5.2]